

Chapter 1

Introduction

Chapter 2

Basic Definitions

Definition 2.1 (σ algebra). A family $\mathcal{P} \subseteq 2^X$ is a σ -algebra if:

1. $\emptyset \in \mathcal{P}$
2. $\mathcal{P}$ is closed under complement: $A \in \mathcal{P} \Leftrightarrow A^c \in \mathcal{P}$
3. $\mathcal{P}$ is closed under **countable** union: $A_n \in \mathcal{P} \Rightarrow \bigcup_{n \in \mathbb{N}} A_n \in \mathcal{P}$

Remark 2.1. Is there any relation between a σ -algebra and a algebra over a ring?

Exercise 2.1 (Closed under countable intersection). Show $\mathcal{P}$ is closed under countable intersection.

$$\bigcap_{n \in \mathbb{N}} A_n \in \mathcal{P}$$

Proof: $A_n \in \mathcal{P} \Rightarrow A_n^c \in \mathcal{P}$. Then

$$\bigcup_{n \in \mathbb{N}} A_n^c \in \mathcal{P} \xrightarrow{\text{De Morgan's Laws}} \left(\bigcap_{n \in \mathbb{N}} A_n \right)^c \in \mathcal{P} \Rightarrow \bigcap_{n \in \mathbb{N}} A_n \in \mathcal{P}$$

□

Definition 2.2 (Outer Measure). A function $\mu : 2^X \rightarrow [0, \infty]$ is an **outer measure** if:

1. $\mu(\emptyset) = 0$
2. $\mu\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i)$ for any $(A_i)_{i \geq 1}$ in 2^X
3. $B \subseteq A \subseteq X \Rightarrow \mu(B) \leq \mu(A)$

Example 2.1. The counting measure $\mu(A) = |A|$ (number of elements in A) is an outer measure on any set X .

Remark 2.2. For which $B \subseteq X$ does $\mu(A) = \mu(A \cap B) + \mu(A \setminus B)$ hold for all $A \in 2^X$? When can B “nicely separate” an arbitrary test set?

Definition 2.3 (μ -Measurable Set). $B \subseteq X$ is μ -measurable if $\forall A \in 2^X$,

$$\mu(A) = \mu(A \cap B) + \mu(A \setminus B).$$

We write $\Sigma(\mu) := \{A \in 2^X : A \text{ is } \mu\text{-measurable}\}.$

Definition 2.4 (Restriction). For $C \subseteq X$, the **restriction** $\mu|_C : 2^X \rightarrow [0, \infty]$ is defined by

$$\mu|_C(A) = \mu(A \cap C).$$

Theorem 2.5. Let μ be an outer measure on X and let $B, C \subseteq X$.

1. $B \subseteq C \Rightarrow \mu(B) \leq \mu(C)$.
2. B is μ -measurable $\Leftrightarrow B^c := X \setminus B$ is μ -measurable.
3. $\mu(B) = 0 \Rightarrow B$ is μ -measurable.
4. X and $\emptyset$ are μ -measurable.
5. B μ -measurable $\Rightarrow B$ is $\mu|_C$ -measurable.

Definition 2.6 (Measure). Let $\mathcal{P}$ be a σ -algebra on X . A function $\mu : \mathcal{P} \rightarrow [0, \infty]$ is a **measure** if:

1. $\mu(\emptyset) = 0$
2. $\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$ whenever $(A_i)_{i \geq 1} \subseteq \mathcal{P}$ with $A_i \cap A_j = \emptyset$ for all $i \neq j$ (countable additivity).

Example 2.2. Given a measure μ on $(X, \mathcal{P})$, define

$$\rho(A) = \inf\{\mu(B) : B \in \mathcal{P}, A \subseteq B\}.$$

Then ρ is an outer measure on X .

2.1 Algebraic Perspective:

One can take a purely algebraic perspective to this discussion of sigma algebras and measures. Let X be a set. Note that 2^X is canonically bijective to $\text{Hom}(X, \mathbb{Z}_2)$. Furthermore, $\text{Hom}(X, \mathbb{Z}_2)$ is an algebra over $\mathbb{Z}_2$ and the addition and multiplication operations correspond to set theoretic operations on elements of 2^X .

Consider the following:

$$\begin{aligned} 1_A + 1_B &= 1_{A \Delta B} \\ 1_A 1_B &= 1_{A \cap B} \\ 1_A + 1_B + 1_{A \cap B} &= 1_{A \Delta B} + 1_{A \cap B} = 1_{(A/B \cup B/A) \Delta A \cap B} \\ &= 1_{A \cup B} \\ 1_X + 1_A &= 1_{A^c} \end{aligned}$$

There is a natural notion of ordering that we can also put on $\text{Hom}(X, \mathbb{Z}_2)$,

$$A \subset B \Rightarrow 1_A 1_B = 1_A \Rightarrow 1_A \leq 1_B.$$

Every element of $\text{Hom}(X, \mathbb{Z}_2)$ is idempotent ($f^2 = f$, since $0, 1 \in \mathbb{Z}_2$ are the only idempotents of $\mathbb{Z}_2$ and squaring is pointwise). A commutative ring in which every element is idempotent is called a **Boolean ring**, so the dictionary above is really the statement that 2^X , with $\cup, \cap, \Delta, (\cdot)^c$, is (canonically) a Boolean ring. This is the starting point of **Stone's representation theorem**: every Boolean ring arises this way, as a ring of subsets of some set.

Remark 2.3. Ring addition $A + B$ is symmetric difference, **not** union, so union is not the additive structure of the ring. It is instead the derived **join**

$$A \vee B := A + B + AB,$$

as one checks on indicators: a point in exactly one of A, B contributes $1 + 0 + 0$ or $0 + 1 + 0$, a point in both contributes $1 + 1 + 1 = 1$, and a point in neither contributes 0 — matching $1_{A \cup B}$ in every case.

Furthermore, we can add the σ portion of the vocabulary to $\text{Hom}(X, \mathbb{Z}_2)$ by allowing sums over a countable number of elements. Some care is needed here: an infinite ring-sum $\sum_{i=1}^{\infty} A_i$ is not **a priori** meaningful, since $\mathbb{Z}_2$ carries no topology to make such a sum converge. It becomes meaningful exactly when the A_i are pairwise **orthogonal**, $A_i A_j = 0$ for $i \neq j$ (i.e. pairwise disjoint) — then at each $x \in X$ at most one term is nonzero, so the sum is well-defined pointwise, and orthogonality forces

$$\sum_{i=1}^{\infty} A_i = \bigvee_{i=1}^{\infty} A_i$$

(the sum agrees with the union). With this understanding, consider the following substructure.

Definition 2.7. $\mathcal{P} \subseteq \text{Hom}(X, \mathbb{Z}_2)$ is a σ -algebra if $\mathcal{P}$ is a subalgebra closed under countable orthogonal sums: whenever $(A_n)_{n \geq 1} \subseteq \mathcal{P}$ satisfy $A_n A_m = 0$ for $n \neq m$,

$$\sum_{n=1}^{\infty} A_n \in \mathcal{P}.$$

Now to define a notion of outer measure.

Definition 2.8 (Outer Measure). $\mu : \text{Hom}(X, \mathbb{Z}_2) \rightarrow [0, \infty]$ is an **outer measure** if

1. $\mu(0) = 0$.
2. $\mu\left(\bigvee_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i)$ for any $(A_i)_{i \geq 1}$ — subadditivity. Note the left side is the **join** $\bigvee$, not the ring sum: the A_i need not be orthogonal.
3. $AB = A \Rightarrow \mu(A) \leq \mu(B)$ — monotonicity, i.e. $A \leq B \Rightarrow \mu(A) \leq \mu(B)$.

Definition 2.9 (μ -Measurable Set). $B \in \text{Hom}(X, \mathbb{Z}_2)$ is **μ -measurable** if for all A ,

$$\mu(A) = \mu(AB) + \mu(A(1+B)).$$

Here $1+B$ is the complement of B (from $1_X + 1_A = 1_{A^c}$ above), so $A(1+B)$ is the indicator of $A \setminus B$, and the right-hand side is a genuine ring sum since AB and $A(1+B)$ are orthogonal.

The notion of a measure in this context is extremely natural, though it is worth being precise about what kind of map it is:

Definition 2.10 (Measure). Let $\mathcal{P}$ be a σ -algebra in $\text{Hom}(X, \mathbb{Z}_2)$. A map $\mu : \mathcal{P} \rightarrow [0, \infty]$ is a **measure** if $\mu(0) = 0$ and, for every pairwise orthogonal family $(A_i)_{i \geq 1} \subseteq \mathcal{P}$,

$$\mu\left(\sum_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

Remark 2.4. A measure is **not** a ring homomorphism $\mathcal{P} \rightarrow [0, \infty]$ — it need not (and generally does not) respect multiplication, i.e. $\mu(AB) \neq \mu(A)\mu(B)$ in general. It is instead an **additive valuation**: a map turning orthogonal ring-sums (which, by the remark above, coincide with joins/unions) into honest sums of real numbers. A Boolean ring equipped with such a valuation is called a **measure algebra**. This algebraic reformulation is not merely decorative: the **Loomis–Sikorski theorem** says every σ -complete Boolean algebra carrying a measure is, up to the ideal of null sets, a quotient of a genuine measure space (Y, Σ, ν) — so “measure theory on abstract Boolean rings” and “measure theory on sets” are the same theory.

Chapter 2.1

References